Theme and chassis layout

antocciphysics | TEACHER KEY | CPA

Learning target: Make a dimensioned layout that supports the circuit and drivetrain.

Suggested first pass: 1, 2, 3, 7. Add independent, mixed and stretch work from the bank as time and student evidence permit. Estimated times are planning estimates, not completion requirements.

01. Bring back an earlier idea (retrieve) *Review: Unit 7, Day 18; 4 course days later.*
24 J; heating or sound are possible.

02. Explain a worked example (worked)
Length account 20=6+5+2+7 cm; real build still needs measured component widths/heights/clearances.

03. Look and explain (model)
Student-specific measured layout with feasible spacing and accessible controls.

04. Build the solution (practice)
18-6-5=7 cm; actual plan must include gaps and component dimensions.

05. Use the idea (reason)
May alter balance, wheel loading or clearance; compare matched runs or inspect/measure loading and straight tracking.

06. Choose a method (mixed) *Review: Unit 7, Day 9; 13 course days later.*
8 A in each branch.

07. Exit check (exit)
Student supplies actual critical dimension and approval status; no automatic approval.

08. STRETCH: Apply it another way (stretch)
Remaining 5 cm after stated lengths/clearances; layouts must avoid overlap and allow wiring/axles; measure actual dimensions.

Teaching note: Daily build worksheet for chassis planning and early assembly. Use the classroom’s current step sequence.

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01. Bring back an earlier idea (retrieve) *Review: Unit 7, Day 18; 4 course days later.*
$|emf|=N|\Delta\Phi|/\Delta t=0.6$ V.

02. Explain a worked example (worked)
Length account 20=6+5+2+7 cm; real build still needs measured component widths/heights/clearances.

03. Represent and interpret (model)
Student-specific measured layout with feasible spacing and accessible controls.

04. Solve and justify (practice)
55 mm remains; .180 m.

05. Analyze the reasoning (reason)
May alter balance, wheel loading or clearance; compare matched runs or inspect/measure loading and straight tracking.

06. Choose a method (mixed) *Review: Unit 7, Day 9; 13 course days later.*
Parallel 6 Ω; total 14 Ω; supply current 2 A.

07. Exit check (exit)
Height/thickness interference may be hidden; include axle height, gear clearance or battery height.

08. STRETCH: Apply it another way (stretch)
40 mm remains. Gear radius, axle height, thickness or battery height may constrain side view.

Teaching note: Daily build worksheet for chassis planning and early assembly. Use the classroom's current step sequence.